

VAIKE-MAARJA, Estonia

WMO: 261410

Lat: 59.1414N

Lon: 26.2308E

Elev: 401

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.0	-0.5	-10.3	3.2	-5.4	-4.7	4.3	0.0	22.9	30.8	21.2	30.5	4.2	120	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	80.4	66.1	77.5	64.6	74.2	62.9	68.4	76.9	66.3	74.4	64.4	71.8	8.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.8	72.6	63.3	88.4	70.4	61.2	81.9	68.3	32.9	76.7	31.2	74.6	29.7	72.0	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.6	16.7	DB	-11.7	84.2	8.0	3.1	-17.5	86.4	-22.2	88.2	-26.7	89.9	-32.5	92.2	(4)
(5)			WB	-12.1	71.4	7.9	2.1	-17.8	73.0	-22.4	74.2	-26.8	75.4	-32.5	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	22.1	21.9	28.1	40.2	50.1	57.5	62.9	60.4	52.1	41.5	32.9	26.7		
	DBStd	16.45	10.80	11.14	8.25	7.59	7.85	6.00	5.69	5.42	5.91	6.97	7.68	9.82		
	HDD50	4297	864	788	679	308	95	7	0	1	47	273	514	721		
	HDD65	8669	1329	1208	1144	744	466	237	110	164	387	730	964	1186		
	CDD50	1184	0	0	0	13	98	232	400	322	110	9	0	0		
	CDD65	80	0	0	0	0	4	11	45	20	0	0	0	0		
	CDH74	723	0	0	0	1	39	112	388	180	3	0	0	0		
	CDH80	104	0	0	0	0	4	6	70	24	0	0	0	0		
	WSAvg	8.3	9.2	9.2	8.9	8.3	7.8	7.8	6.8	6.9	7.5	8.4	9.2	10.0		
	PrecAvg	26.8	1.9	1.5	1.4	1.3	1.9	3.0	3.1	3.5	2.3	2.8	2.3	2.0		
(15) Precipitation	PrecMax	35.5	3.7	3.7	2.5	2.5	4.4	5.7	6.1	7.4	4.1	5.0	4.2	3.9		
	PrecMin	17.4	0.4	0.3	0.4	0.4	0.3	0.6	0.3	0.2	0.6	0.4	0.4	1.0		
	PrecStd	4.1	0.7	0.8	0.6	0.6	1.0	1.4	1.4	2.0	1.1	0.9	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	40.3	52.6	69.7	78.9	80.8	85.5	82.8	73.4	59.0	48.8	46.2		
		MCWB	39.5	37.8	43.6	53.5	63.0	65.3	69.0	66.3	61.1	54.1	47.2	44.4		
	2%	DB	37.7	37.9	45.0	63.9	73.1	77.7	81.9	78.7	68.0	55.5	46.9	43.0		
		MCWB	36.8	36.4	39.1	51.2	59.9	63.6	67.4	64.9	59.4	52.9	45.7	41.7		
	5%	DB	35.7	36.3	40.7	58.5	69.0	73.4	78.4	75.1	64.6	53.4	45.0	40.4		
		MCWB	34.8	35.0	36.6	47.8	56.8	61.5	66.2	63.1	57.3	51.5	43.9	39.3		
	10%	DB	34.2	34.6	37.6	53.5	64.9	69.7	75.0	71.6	61.7	51.2	43.0	37.8		
		MCWB	33.4	33.6	34.7	45.1	53.8	59.4	64.3	61.9	56.2	49.3	42.1	36.8		
	0.4%	WB	39.8	38.6	44.1	56.0	65.9	68.4	71.6	70.1	62.8	55.4	47.6	44.8		
		MCDB	40.6	39.4	50.8	66.6	74.8	77.2	81.3	78.4	69.3	57.2	48.3	46.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	36.7	39.7	51.8	61.9	65.8	69.5	66.6	60.5	53.4	45.9	41.9		
		MCDB	37.4	37.7	43.8	61.3	70.0	73.6	78.4	74.8	65.7	54.8	46.8	42.8		
	5%	WB	35.0	35.1	37.3	49.0	58.6	63.1	67.4	64.7	58.8	51.7	44.0	39.4		
		MCDB	35.5	36.0	39.8	57.6	66.5	70.4	75.4	72.0	63.2	53.3	44.9	40.2		
	10%	WB	33.5	33.7	35.4	45.8	55.2	60.6	65.5	62.9	57.0	49.7	42.1	36.9		
		MCDB	34.1	34.4	37.2	52.3	62.8	68.0	73.3	70.0	60.5	51.1	42.9	37.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.2	12.0	15.9	18.7	17.0	17.0	17.1	14.1	9.8	6.7	7.1		
		MCDBR	6.1	7.6	15.1	24.8	24.0	22.9	21.9	22.9	18.4	11.6	7.6	6.4		
		MCWBR	6.1	6.9	10.3	13.8	12.8	11.6	10.1	11.0	10.7	8.8	7.2	6.3		
		MCDBR	6.2	7.2	12.7	22.6	21.4	19.2	19.6	19.3	16.0	10.2	7.4	6.0		
	5% WB	MCWBR	6.2	6.8	9.2	13.3	12.2	11.1	9.9	10.5	10.3	8.3	7.1	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.301	0.327	0.367	0.363	0.361	0.383	0.382	0.350	0.330	0.305	0.258		
	taud		2.111	2.138	2.189	2.300	2.391	2.426	2.382	2.384	2.463	2.408	2.205	2.080		
	Ebn at Noon		165	221	254	265	275	277	267	258	249	215	151	124		
	Edh at Noon		15	25	32	34	34	33	34	31	25	19	14	11		
(44) All-Sky Solar Radiation	RadAvg		88	279	741	1225	1700	1754	1702	1326	819	348	99	53		
	RadStd		15	51	100	141	180	141	157	134	100	49	15	8		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	-279	-292	N/A	N/A

Nomenclature: See separate page